Prateek Kumar

Satna, Madhya Pradesh
+91-7898853505 | prateekk02.kumar@gmail.com | LinkedIn | GitHub | LeetCode | Portfolio

Experience

BITS Pilani (Goa) **August 2024 - January 2025**
AIML Intern *Remote*

- Accelerated CFD simulation result output from **40** minutes to **4** minutes (**10x** speedup) using deep learning
- Developed a deep learning ANN model that achieved **98%** accuracy.
- Optimized the ANN algorithm to efficiently process **50,000+** data points and improved computational performance.
- Created **12+** data visualizations to track key performance metrics and support data-driven decision-making.
- Collaborated with cross-functional teams to develop real-time API analytics solutions.
- Technologies Used: Python, NumPy, Pandas, Matplotlib, TensorFlow, Scikit Learn, Deep Learning (ANN Architecture).

Projects

YouTube Chatbot - Real-time RAG Application — GitHub Link

- Built a Streamlit chatbot that provides real-time Q&A on any YouTube video by extracting and querying its transcript.
- Automated transcript extraction and chunking, processing lengthy videos for precise retrieval without manual effort.
- Integrated OpenAI embeddings, ChatModel, LangChain, and FAISS for sub-second answers on 10k+ tokens.
- Technologies Used: Python, Streamlit, LangChain, Vector Database(FAISS, PineCone), OpenAI API, RAG.

Content Catalyst - An AI SaaS — GitHub Link — Live Demo

- Developed an AI-powered platform for content creators to streamline media editing and enhancement.
- Integrated Clerk auth API, trimming authentication code from **300–600** to just **20–40** LOC (**-90 %**).
- Set up FAL AI webhooks to deliver inference results in under **500 s**, eliminating polling and simplifying data flow.
- Enhanced content delivery with Cloudinary's video compression, achieving a **98%** reduction in video size.
- Technologies Used: Next.js, TypeScript, Fal AI API, Clerk, Cloudinary, Prisma, PostgreSQL, Docker Compose, Vercel

End-to-End Twitter Depression Prediction System — GitHub Link

- Designed an API-driven depression-risk system that analyzes Twitter feeds with **90%** accuracy.
- Built an NLP pipeline—spaCy preprocessing + TensorFlow Bi-LSTM that lifted baseline F1 by **15%** on a held-out set.
- Automated Twitter data acquisition via Selenium to continuously supply training datasets.
- Technologies Used: Python, Flask-RESTful, Selenium (Web Driver), React.js, spaCy, Scikit-learn, TensorFlow

Education

Lakshmi Narain College of Technology, Bhopal **August 2021 - May 2025**
Bachelor of Technology in Computer Science and Engineering (CGPA: 8.75/10) *Madhya Pradesh, India*

- Relevant Coursework: DSA, DBMS, OOPs, Operating System, Machine Learning, Computer Vision.
- First Division With Honours

St. Michael's Hr. Sec. School, Satna **May 2020**
12th Grade (CBSE) (82.4%) *Madhya Pradesh, India*

- Relevant Coursework: Physics, Chemistry, Mathematics, Computer Science (C++).

Technical Skills

- **Programming Languages:** C++, Python, JavaScript, TypeScript, SQL
- **Full-Stack Development:** Next.js, React.js, Express.js, Node.js, Django, Pydantic
- **Databases:** PostgreSQL, MongoDB, Prisma
- **DevOps & Cloud:** Git, GitHub, Docker, Docker Compose, Cloudflare Workers, AWS (S3, EC2), Nginx Reverse Proxy
- **AI/ML:** TensorFlow, Scikit-learn, Seaborn, spaCy, Deep Learning, Computer Vision, NLP, LangChain, RAG

Achievements and Certifications

- Led teams in national-level hackathons such as Hackhive, Hackwave, and Flipkart Grid.
- Spearheaded a team of 4-5 members as GDSC AIML Lead, driving AI/ML initiatives and delivering 3+ events.
- Achieved an All India Rank of **6339** in GATE-DA 2024. **Link**
- Secured a position in the top **5**% merit list for NPTEL Discrete Mathematics Course by IIT Ropar.